

Technical Assessment Write-up

Jeroen Buil

Task 2 – PsychoPy: Posner cueing task

Design choices:

I designed the task (Posner_Cueing_Task.psyexp) in PsychoPy Builder as the official documentation noted potential problems when running on Linux (typing this on CachyOS). To avoid unnecessary delays I opted for designing the experiment this way.

Task:

- **Inter Trial Interval (ITI) added**
Instructions had no mention of an ITI, but to prevent the next trial immediately starting after the previous one ended. I added a ITI period with a jittered length of 0.8-1.2 seconds.
- **Start + End screen added**
Instructions had no mention of this either, but having instructions before the experiment starts and one indicating the end, makes it much more clear to the user (and developer) when the experiment started and ended.
- **80% valid ratio** is achieved by having 4-to-1 valid-to-invalid ratio of columns in the conditions file. I.e. each invalid cueSide and targetSide (left/right) combination had only 1 row, while every valid combination has 4 rows.
For each cueType (endo/exogenous) you then get 10 entries (4+4+1+1) of which 8 (4+4) valid = 80%
- **Cue image file path are read from the conditions file.** These paths could also have been coded into the routine image box. This is developer preference in my eyes. I chose this, as it allows a new user of this experiment to more easily adjust the cue images without having to open the .psyexp file and writing if/else code statements.
- **Loop only covers trial and ITI routines**, the setup, instructions and end message should only appear before the looping of the trials. The number of trial repetitions is set with the nRep (Number of Repetitions) setting inside the loop.
- **nRep = 1.** There are 20 different stimuli in total, and they currently aren't repeated (to keep the experiment short for testing). If desired, the number of repetitions can be increased in the Flow loop 'trials'. E.g. nRep = 5 => 100 trials.

Triggers:

How:

- A trigger is send/logged every time a new trial phase onsets (i.e. fix, cue, target) or when a button is pressed (the response)
- Triggers are send both written in the experiment .csv file as in a separate _triggers.csv file. The latter is meant to mimic what would be send over the EEG/MEG marker

channel. These normally work with numeric codes that each have a meaning (see: `trigger_codes.csv`). E.g.

- o 0 Fixation onset
- o 11 Cue onset left
- o 12 cue onset right
- o 30 incorrect response
- o 31 correct response
- o Etc.

Note that I am also writing the actual response label as string, to make it more readable for this assignment. In a real experiment sending just the timestamp and numeric code would be sufficient.

Where:

- Triggers should be fired on the same frame the stimulus appears on the screen. The code for this is housed within the 'Each Frame' tab in the code block of the corresponding routine. It checks against the internal clock that drives the stimulus onsets and ensures these align.
The only deviation from the each Frame this is for the Intertrial Interval (ITI) where it is set in Begin Routine, which simply runs at the start of this specific routine. It sends the ITI_ONSET trigger when to mark the onset and then nothing happens, nor needs to be monitored, until the next trial starts.
- The actual trigger call `send_trigger()` is defined in the setup routine that runs at the start of the experiment. It defines the code dictionary as well as other global variables used throughout the experiment.

Additional Notes:

- In a real experiment the EEG/MEG data quality might be poor around the response due to movement artefacts from button presses. For this reason, this specific task seems to be more suitable for analysing the brain's response between cue and target and not between target and response.
- The layout of the visual elements has only been tested on a 4K monitor. Good practice would be to run the pilot again on the intended monitor to ensure the elements appear where they should in the intended size.